

Sai Rama Chandra Mamidala

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Career Objective

Computer Science undergraduate at GITAM University with hands-on experience in backend development, AI/ML systems, and full-stack platforms. Passionate about building scalable, production-ready software and solving complex architectural problems.

Education

B.Tech in Computer Science Engineering (Pursuing) — GITAM University, Hyderabad CGPA: 8.14 / 10	2023 – 2027
Intermediate (TS Board) — Sri Chaitanya Junior College 91.4%	2021 – 2023
Secondary Education — Sri Chaitanya School 82.2%	2021

Hackathon Experience

Codryx AI Hackathon — Hyderabad 2026

Project: QuoteAI — Autonomous Insurance Underwriting System

- **6-agent ML pipeline** replacing manual underwriting with automated risk modeling and fraud detection.
- Implemented **XGBoost** for conversion prediction, **Random Forest** for risk scoring, and **Isolation Forest** for anomaly-based fraud detection.
- Secured an **internship offer** from Codryx AI based on technical excellence and innovative implementation.

Projects

Ivana Designers Studio — Fashion Rental & Retail SaaS Platform 2026

- Architected a full-stack SaaS platform with **React storefront**, **React/TypeScript admin portal**, and **Strapi 5 REST API** on Supabase PostgreSQL.
- Implemented **PostgreSQL advisory locks** to prevent double-booking race conditions; integrated **Paytm payment gateway** with idempotent refund processing.
- Designed **7 custom n8n automation workflows**; built **Playwright E2E test suite** across 9 spec files with live database verification.
- Conducted a full **security audit remediating 27+ vulnerabilities** including XSS vectors, auth misconfigurations, and silent data corruption bugs.

Autonomous Quote Agent — Multi-Agent Insurance System 2026

- **6-agent ML pipeline** for automated risk modeling and fraud detection in motor insurance underwriting.
- Used **XGBoost**, **Random Forest**, and **Isolation Forest** for conversion prediction, risk scoring, and anomaly detection.

DevOps YAML Automation Platform 2025

- Built a **FastAPI backend** with key-path allow-list policies and **GitHub API** integration for automated PR generation across multiple repositories.
- Developed an operator dashboard in **Next.js/TypeScript** for visual diff previews and execution commands.

Ram Jam — Real-Time Collaborative Music Player 2024

- Built a low-latency web app for **synchronized YouTube playback** using **Node.js**, **Express**, and **Socket.IO** with resilient WebSocket communication.

PTCE — Predictive Traffic Control Engine

In Development

- Building a **real-time predictive traffic intelligence platform** for monitoring and forecasting urban congestion using event-driven systems.
- Designing **streaming data pipelines** and **graph-based traffic modeling** to analyze flow across complex city networks.
- Developing **predictive analytics modules** to forecast short-term congestion risks and support smart city traffic management.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C

Backend: Node.js, Express, FastAPI, Strapi 5, Socket.io

Frontend: React, Next.js, Vite, Tailwind CSS, HTML5, CSS3

AI / ML: Scikit-learn, XGBoost, Random Forest, Isolation Forest

Databases: PostgreSQL (Supabase), MySQL

Automation: n8n (multi-workflow orchestration, webhook pipelines, cron)

Testing & QA: Playwright (E2E), Vitest, race condition testing, live DB verification

DevOps & Cloud: Git, Docker, Linux, Nginx, PM2, Vercel, Railway, Hetzner VPS, DigitalOcean

Leadership & Interests

Competitive Debating

- Active participant in competitive debating and philosophical discourse.

Open-Source & Music

- Pursues music production and sound engineering alongside active open-source software development.